10.2 lubridate: Dates and Times

A string that looks like a date is still text. Parse it, then pull out pieces, add calendar time, and keep a duration distinct from a period.

1. A string is not a date

A column can look like a date and still be text. Text does not know that November comes after October, and you cannot subtract two strings to get a number of days.

```
x <- "2026-11-11"
class(x)
```

```
## [1] "character"
```

Turn the string into a date or a datetime before you filter, subtract, or plot. Three tools do that conversion.

1. Base R: `as.Date()`, `as.POSIXct()`, `strptime()`.
2. readr: `parse_date()`, `parse_datetime()`, `parse_time()`.
3. lubridate: `ymd()`, `mdy()`, `dmy()`, `ymd_hms()`, and the helpers that come after the parse.

All three can turn a string into a date. lubridate is the one that also makes the next steps short: pull out the month, add two calendar months, change the time zone.

tidyverse attaches readr and lubridate. Load lubridate yourself so the parsers are obvious.

```
library(tidyverse)
library(lubridate)
```

---

2. Date types in R

A date type is either a moment on the calendar or a length of time. The two tables below are the ones you will meet in this course.

A moment: a day, an instant, or a clock time

Type	What it stores	Class
Date	A calendar day, with no time of day	`Date`
Date-time	One instant, stored as seconds since 1970-01-01 00:00:00 UTC	`POSIXct`, `POSIXt`
Date-time, as pieces	The same kind of instant, stored as a list of year, month, day, hour, and the rest	`POSIXlt`, `POSIXt`
Time of day	A clock time with no date, such as 14:30:00	`hms`, `difftime`

`Date`, `POSIXct`, and `POSIXlt` are base R. `hms` comes from the `hms` package, which `readr` loads. For a column in a data frame, use `POSIXct`. `strptime()` returns `POSIXlt`.

```
class(as.Date("2026-11-11"))
class(as.POSIXct("2026-11-11 14:30:00", tz = "UTC"))
class(strptime("2026-11-11 14:30:00", format = "%Y-%m-%d %H:%M:%S", tz = "UTC"))
class(parse_time("14:30:00"))
```

```
## [1] "Date"
## [1] "POSIXct" "POSIXt"
## [1] "POSIXlt" "POSIXt"
## [1] "hms"      "difftime"
```

```
ymd("2017-01-31")
ymd_hms("2017-01-31 20:11:59")
```

```
## [1] "2017-01-31"
## [1] "2017-01-31 20:11:59 UTC"
```

The first is a `Date`. The second is a `POSIXct`, printed in UTC.

A length of time

These types do not name a day. They say how long something lasts.

Type	What it stores	Class	How you get one
Difference	How far apart two dates are	`difftime`	`end - start`
Duration	An exact number of seconds	`Duration`	`ddays(1)`
Period	A calendar unit. A day can be 23 or 25 hours when the clocks change.	`Period`	`days(1)`, `months(1)`, `years(1)`
Interval	The span between two instants	`Interval`	`interval(start, end)`

`difftime` is base R. Duration, period, and interval are lubridate. Section 7 subtracts two dates. Section 8 shows why a duration and a period can land on different days.

```
class(as.Date("2026-11-11") - as.Date("2026-11-01"))
class(ddays(1))
class(days(1))
class(interval(ymd("2026-11-01"), ymd("2026-11-11")))
```

```
## [1] "difftime"
## [1] "Duration"
## attr(,"package")
## [1] "lubridate"
## [1] "Period"
## attr(,"package")
## [1] "lubridate"
## [1] "Interval"
## attr(,"package")
## [1] "lubridate"
```

---

3. Base R

No extra package. You tell R the layout of the string with `format`.

`as.Date()`

Use this for a calendar date.

```
x <- "2026-11-11"
d <- as.Date(x)
d
class(d)
```

```
## [1] "2026-11-11"
## [1] "Date"
```

ISO text, year-month-day, is the default. Any other layout needs `format`.

```
as.Date("11/25/2026", format = "%m/%d/%Y")
as.Date("25-Nov-2026", format = "%d-%b-%Y")
```

```
## [1] "2026-11-25"
## [1] "2026-11-25"
```

Symbol	Meaning	Example
`%Y`	Four-digit year	2026
`%y`	Two-digit year	26
`%m`	Month number	11
`%d`	Day of the month	25
`%b`	Abbreviated month	Nov
`%B`	Full month	November
`%H`	Hour, 00–23	14
`%I`	Hour, 01–12	02
`%p`	AM or PM	PM
`%M`	Minute	30
`%S`	Second	45

`%m` is the month. `%M` is the minute.

`as.POSIXct()`

Use this when the string has a date and a time. `tz` is the time zone.

```
dt <- as.POSIXct(
  "2026-11-11 14:30:00",
  format = "%Y-%m-%d %H:%M:%S",
  tz = "UTC"
)
dt
class(dt)
```

```
## [1] "2026-11-11 14:30:00 UTC"
## [1] "POSIXct" "POSIXt"
```

`strptime()`

`strptime()` also parses a datetime, and it returns a `POSIXlt`.

```
lt <- strptime(
  "11/25/2026 14:30",
  format = "%m/%d/%Y %H:%M",
  tz = "UTC"
)
lt
class(lt)
```

```
## [1] "2026-11-25 14:30:00 UTC"
## [1] "POSIXlt" "POSIXt"
```

`POSIXct` stores the instant as one number of seconds. `POSIXlt` stores a list of pieces: year, month, day, hour, and the rest. For a column in a data frame, use `POSIXct`.

`format()` goes back to text

`format()` prints a date in a layout you choose. The result is a character string, not a `Date`.

```
d <- as.Date("2026-11-11")
format(d, "%B %d, %Y")
format(d, "%Y")
class(format(d, "%Y"))
```

```
## [1] "November 11, 2026"
## [1] "2026"
## [1] "character"
```

---

4. readr

readr parses character vectors, and it records the values it could not parse.

Function	Input	Class
`parse_date()`	a calendar date	`Date`
`parse_datetime()`	a date and a time	`POSIXct`
`parse_time()`	a time of day	`hms`

```
parse_date("2026-11-11")
parse_date("11/25/2026", format = "%m/%d/%Y")

parse_datetime("2026-11-11 14:30:00")

parse_time("14:30:00")
parse_time("2:30pm")
```

```
## [1] "2026-11-11"
## [1] "2026-11-25"
## [1] "2026-11-11 14:30:00 UTC"
## 14:30:00
## 14:30:00
```

`parse_datetime()` prints the result in UTC unless the string or the locale says otherwise. `parse_time()` is the right tool for a column that is only a clock time, such as a store opening.

A bad value becomes `NA`, and readr keeps a record of why.

```
dates <- parse_date(c("2026-11-11", "2026-13-25", "2026-12-01"))
dates
problems(dates)
```

```
## Warning: 1 parsing failure.
## row col   expected     actual
##   2  -- valid date 2026-13-25
## [1] "2026-11-11" NA           "2026-12-01"
## # A tibble: 1 × 4
##     row   col expected   actual
##   <int> <int> <chr>      <chr>
## 1     2    NA valid date 2026-13-25
```

Month 13 is not a month. `problems()` shows the row and the string that failed. That is the reason to parse with readr while you are still importing.

You can parse during `read_csv()` instead of converting a character column later. Put the course CSV files in a `data/` folder of your project.

```
read_csv(
  "data.csv",
  col_types = cols(
    date = col_date(format = "%m/%d/%Y"),
    time = col_time(),
    .default = col_guess()
  )
)
```

`wmata_ridership.csv` has a `Date` column of ISO text and a `Total` column of daily Metro rail rides. lubridate parses that column in the next section. `col_date()` would do the same job inside `read_csv()`.

Base R and readr both turn strings into dates. readr is the one that sets column types at import and keeps a table of the failures.

---

5. lubridate parsers

The function name is the order of the pieces. `y` is year, `m` is month, `d` is day. The separators can vary.

```
ymd("2026-11-25")
mdy("11/25/2026")
dmy("25/11/2026")

ymd("2017-01-31")
mdy("January 31st, 2017")
dmy("31-Jan-2017")
```

```
## [1] "2026-11-25"
## [1] "2026-11-25"
## [1] "2026-11-25"
## [1] "2017-01-31"
## [1] "2017-01-31"
## [1] "2017-01-31"
```

All of those are `Date` objects. The same string, three ways:

```
x <- "11/25/2026"
as.Date(x, format = "%m/%d/%Y")
parse_date(x, format = "%m/%d/%Y")
mdy(x)
```

```
## [1] "2026-11-25"
## [1] "2026-11-25"
## [1] "2026-11-25"
```

Tool	Call	Result
Base R	`as.Date(x, format = "%m/%d/%Y")`	`2026-11-25`
readr	`parse_date(x, format = "%m/%d/%Y")`	`2026-11-25`
lubridate	`mdy(x)`	`2026-11-25`

`mdy()` does not need `"%m/%d/%Y"`. The name of the function is the format.

Add `_hms`, `_hm`, or `_h` when the string has a clock time. Pass `tz` when you know the zone.

```
ymd_hms("2026-11-11 14:30:45", tz = "UTC")
ymd_hm("2026-11-11 14:30", tz = "UTC")
mdy_hms("11/11/2026 14:30:45", tz = "UTC")
dmy_hm("11 November 2026 14:30", tz = "UTC")

ymd_hms("2017-01-31 20:11:59")
mdy_hm("01/31/2017 08:15")
```

```
## [1] "2026-11-11 14:30:45 UTC"
## [1] "2026-11-11 14:30:00 UTC"
## [1] "2026-11-11 14:30:45 UTC"
## [1] "2026-11-11 14:30:00 UTC"
## [1] "2017-01-31 20:11:59 UTC"
## [1] "2017-01-31 08:15:00 UTC"
```

`make_date()` and `make_datetime()` build a date from numeric columns.

```
make_date(2017, 1, 31)
make_datetime(2017, 1, 31, 20, 11, 59)
```

```
## [1] "2017-01-31"
## [1] "2017-01-31 20:11:59 UTC"
```

Once `wmata_ridership.csv` is in `data/`, parse `Date` with `ymd()`. This chunk is not run here, because that file is not part of the notes.

```
wmata <- read_csv("data/wmata_ridership.csv")
wmata <- wmata |>
  mutate(Date = ymd(Date))
```

---

6. Components

After the parse, lubridate pulls the pieces out as numbers, or as an ordered factor when you ask for a label.

```
d <- ymd("2026-11-11")
year(d)
month(d)
day(d)
month(d, label = TRUE)
wday(d, label = TRUE)
```

```
## [1] 2026
## [1] 11
## [1] 11
## [1] Nov
## 12 Levels: Jan < Feb < Mar < Apr < May < Jun < Jul < Aug < Sep < ... < Dec
## [1] Wed
## Levels: Sun < Mon < Tue < Wed < Thu < Fri < Sat
```

`label = TRUE` returns an ordered factor. `abbr = FALSE` spells the name out. The same extractors work on a datetime.

```
dt <- ymd_hms("2016-07-08 12:34:56")
year(dt)
month(dt)
month(dt, label = TRUE)
day(dt)
wday(dt)
wday(dt, label = TRUE)
hour(dt)
yday(dt)
quarter(dt)
```

```
## [1] 2016
## [1] 7
## [1] Jul
## 12 Levels: Jan < Feb < Mar < Apr < May < Jun < Jul < Aug < Sep < ... < Dec
## [1] 8
## [1] 6
## [1] Fri
## Levels: Sun < Mon < Tue < Wed < Thu < Fri < Sat
## [1] 12
## [1] 190
## [1] 3
```

Function	What it returns
`year()`	year
`month()`	month
`day()`	day of the month
`wday()`	weekday
`yday()`	day of the year
`week()`	week number
`quarter()`	quarter, 1–4
`hour()`	hour
`minute()`	minute
`second()`	second

`wday()` counts Sunday as 1 unless you set `week_start`.

Base R can print the same pieces with `format()`, and the result is text. `year()` returns a number.

```
format(d, "%Y")
class(format(d, "%Y"))
year(d)
class(year(d))
```

```
## [1] "2026"
## [1] "character"
## [1] 2026
## [1] "numeric"
```

You can also replace one component. The other pieces stay.

```
d <- ymd("2026-11-11")
year(d) <- 2027
d
month(d) <- 12
d
```

```
## [1] "2027-11-11"
## [1] "2027-12-11"
```

That assignment can build a date that does not exist, such as February 31. Check the result around month ends and leap days.

On the WMATA file, after the parse in the previous section:

```
wmata |>
  mutate(
    year = year(Date),
    month = month(Date, label = TRUE),
    wday = wday(Date, label = TRUE)
  )
```

---

7. Calendar arithmetic

`days()`, `months()`, and `years()` add calendar units.

```
d <- ymd("2026-11-11")
d + days(10)
d + months(2)
d + years(1)
```

```
## [1] "2026-11-21"
## [1] "2027-01-11"
## [1] "2027-11-11"
```

Subtract two dates to get a difftime.

```
start <- ymd("2026-11-01")
end <- ymd("2026-11-11")
end - start
```

```
## Time difference of 10 days
```

`interval()` stores the two endpoints. `time_length()` turns that span into a number of days.

```
interval(start, end)
time_length(interval(start, end), unit = "days")
```

```
## [1] 2026-11-01 UTC--2026-11-11 UTC
## [1] 10
```

`floor_date()` and `ceiling_date()` snap a date to the start of a week, a month, or a year. That is how you group daily rows into months.

```
d <- ymd("2026-11-11")
floor_date(d, unit = "month")
ceiling_date(d, unit = "month")
```

```
## [1] "2026-11-01"
## [1] "2026-12-01"
```

---

8. Durations, periods, and intervals

The length types from section 2, next to the functions that build them:

Type	What it stores	Class	How you get one
Difference	How far apart two dates are	`difftime`	`end - start`
Duration	An exact number of seconds	`Duration`	`ddays(1)`
Period	A calendar unit. A day can be 23 or 25 hours when the clocks change.	`Period`	`days(1)`, `months(1)`, `years(1)`
Interval	The span between two instants	`Interval`	`interval(start, end)`

A **duration** is an exact number of seconds. `ddays(1)` is always 86,400 seconds. `dyears(1)` is 365.25 of those days.

A **period** is a human unit. `days(1)` is one calendar day, which can be 23 or 25 hours when the clocks change. `years(1)` is one calendar year, 365 or 366 days.

An **interval** is the span between two instants.

```
ddays(1)
days(1)
years(1)

ymd("2016-01-01") + ddays(365)
ymd("2016-01-01") + days(365)
ymd("2016-01-01") + years(1)
ymd("2016-01-01") + dyears(1)
```

```
## [1] "86400s (~1 days)"
## [1] "1d 0H 0M 0S"
## [1] "1y 0m 0d 0H 0M 0S"
## [1] "2016-12-31"
## [1] "2016-12-31"
## [1] "2017-01-01"
## [1] "2016-12-31 06:00:00 UTC"
```

2016 is a leap year, so a calendar year is 366 days. `ddays(365)` and `days(365)` both land on 2016-12-31. `years(1)` lands on 2017-01-01. `dyears(1)` is 365.25 days of seconds, so the result is a datetime six hours before the end of December 31.

```
int <- interval(ymd("2004-01-01"), ymd("2018-12-21"))
int
as.duration(int)
```

```
## [1] 2004-01-01 UTC--2018-12-21 UTC
## [1] "472435200s (~14.97 years)"
```

Use a duration when you want elapsed seconds. Use a period when you want the same date next month. Use an interval when you have two endpoints.

---

9. Time zones

A datetime from `ymd_hms()` with no `tz` is **UTC**.

`with_tz()` keeps the instant and redraws the clock in another zone. `force_tz()` keeps the clock face and changes which instant that clock face means.

```
dt <- ymd_hms("2026-11-11 14:30:00", tz = "UTC")
with_tz(dt, "America/New_York")
force_tz(dt, "America/New_York")
```

```
## [1] "2026-11-11 09:30:00 EST"
## [1] "2026-11-11 14:30:00 EST"
```

14:30 UTC is 09:30 Eastern that day. `force_tz()` leaves the digits 14:30 and labels them Eastern, which is a different instant.

Function	Clock	Instant
`with_tz()`	changes	stays the same
`force_tz()`	stays the same	changes

`OlsonNames()` lists the zone names. `"America/New_York"` is Eastern time, daylight saving included.

The same pair on a datetime that was parsed with no zone, so lubridate stored it as UTC:

```
x <- ymd_hms("2020-03-08 01:30:00")
x
force_tz(x, "America/New_York")
with_tz(x, "America/New_York")
```

```
## [1] "2020-03-08 01:30:00 UTC"
## [1] "2020-03-08 01:30:00 EST"
## [1] "2020-03-07 20:30:00 EST"
```

A clock that was recorded in local time is often stored as UTC by accident. On the morning the clocks spring forward, 2:00 a.m. does not exist, so a span that looks like two hours is one hour of real time. Subtracting datetimes gives a difftime. Ask `as.numeric()` for hours. Dividing by 3,600 is right only when that difftime is still in seconds. For a gap of a couple of hours, R has already chosen hours.

```
toy <- tibble(
  start = mdy_hm(c("3/13/2016 1:30", "3/13/2016 1:30")),
  end   = mdy_hm(c("3/13/2016 3:30", "3/13/2016 3:30"))
)
toy |>
  mutate(
    start_ny = force_tz(start, "America/New_York"),
    end_ny = force_tz(end, "America/New_York"),
    hours_utc = as.numeric(end - start, units = "hours"),
    hours_ny = as.numeric(end_ny - start_ny, units = "hours")
  ) |>
  select(hours_utc, hours_ny)
```

```
## # A tibble: 2 × 2
##   hours_utc hours_ny
##       <dbl>    <dbl>
## 1         2        1
## 2         2        1
```

UTC reports 2 hours. New York reports 1 hour. `force_tz()` is the fix when the numbers on the clock were already Eastern.

---

10. One small data set

Five transactions. `date_string` is text.

```
sales <- tibble(
  customer = c("A", "B", "C", "D", "E"),
  date_string = c(
    "11/01/2026",
    "11/03/2026",
    "11/03/2026",
    "11/08/2026",
    "11/11/2026"
  ),
  amount = c(100, 150, 80, 200, 120)
)

sales <- sales |>
  mutate(
    date = mdy(date_string),
    year = year(date),
    month = month(date),
    weekday = wday(date, label = TRUE)
  )

sales
```

```
## # A tibble: 5 × 7
##   customer date_string amount date        year month weekday
##   <chr>    <chr>        <dbl> <date>     <dbl> <dbl> <ord>
## 1 A        11/01/2026     100 2026-11-01  2026    11 Sun
## 2 B        11/03/2026     150 2026-11-03  2026    11 Tue
## 3 C        11/03/2026      80 2026-11-03  2026    11 Tue
## 4 D        11/08/2026     200 2026-11-08  2026    11 Sun
## 5 E        11/11/2026     120 2026-11-11  2026    11 Wed
```

customer	date	year	month	weekday
A	2026-11-01	2026	11	Sun
B	2026-11-03	2026	11	Tue
C	2026-11-03	2026	11	Tue
D	2026-11-08	2026	11	Sun
E	2026-11-11	2026	11	Wed

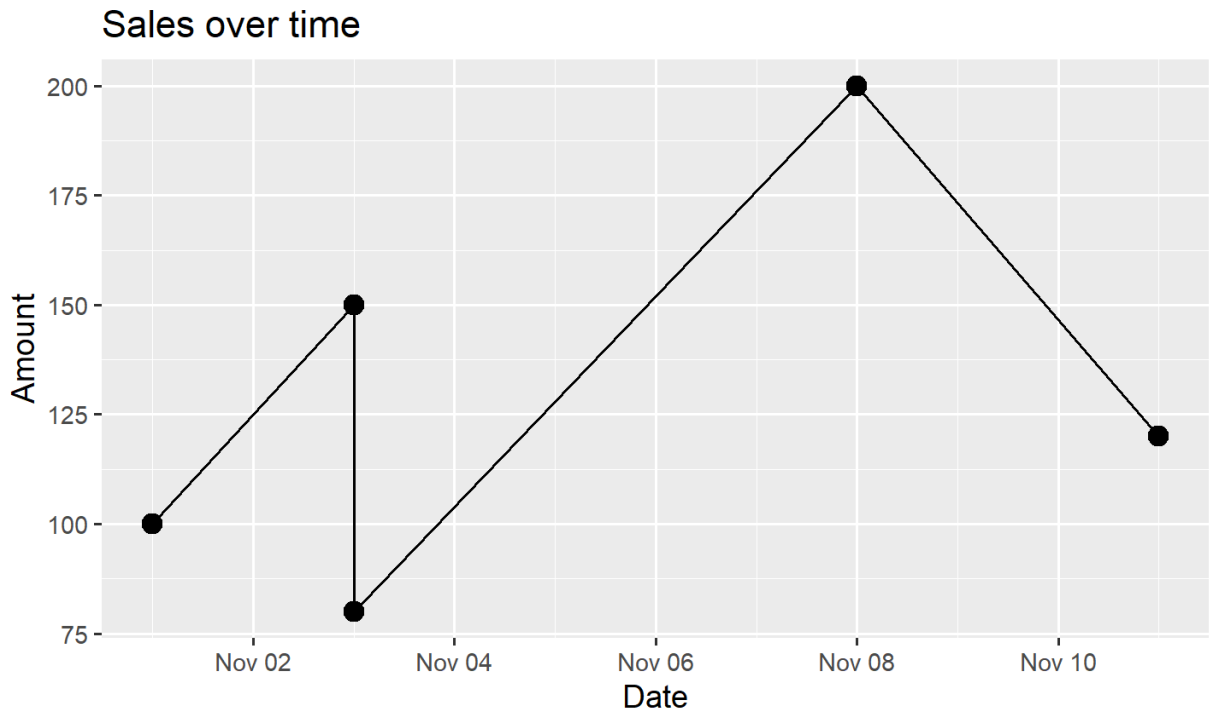
Sum the amount by weekday. The weekday factor puts Sunday first.

```
sales |>
  group_by(weekday) |>
  summarise(total_sales = sum(amount), .groups = "drop")
```

```
## # A tibble: 3 × 2
##   weekday total_sales
##   <ord>         <dbl>
## 1 Sun             300
## 2 Tue             230
## 3 Wed             120
```

Then plot. B and C share November 3, so the line drops vertically between 150 and 80 on that day.

```
sales |>
  ggplot(aes(x = date, y = amount)) +
  geom_point(size = 3) +
  geom_line(group = 1) +
  labs(title = "Sales over time", x = "Date", y = "Amount")
```



The order of the work is: read the rows, parse the date, pull out the pieces you need, then summarise or plot.

---

11. Which tool

Task	Base R	readr	lubridate
Parse a date	`as.Date()`	`parse_date()`	`ymd()`, `mdy()`, `dmy()`
Parse a datetime	`as.POSIXct()`	`parse_datetime()`	`ymd_hms()` and the rest
Parse a time of day	`strptime()` plus a date	`parse_time()`	`hms()`, which is a period
Say how the string is laid out	`%Y-%m-%d` and the rest	the same codes	usually the function name
Year as a number	`format()` returns text		`year()`
Month, weekday	`format()`, `weekdays()`		`month()`, `wday()`
Add a calendar month	you build it		`months()`
Snap to the start of a month	you build it		`floor_date()`
See which rows failed to parse	little built in	`problems()`	a warning, and `NA`
Change the time zone	`tz`, then convert	the locale	`with_tz()`, `force_tz()`

`lubridate::hms()` builds a **period**: 14 hours, 30 minutes, 0 seconds. `readr::parse_time()` builds an **hms** time of day. They can print looking alike. `class()` tells them apart.

Use base R when you are learning the classes and the `%` codes, or when you cannot add a package. Use readr when you are importing a file and you need to know which rows failed. Use lubridate when the layouts vary, or when you need components, calendar arithmetic, grouping by time, or time zones.

They fit together. Import with readr. Manipulate the dates with lubridate.

---

12. Questions

1. Which readr function turns a string that is only a date into a `Date`?
2. Which lubridate function parses `"25/11/2026"`?
3. What is the difference between `format(d, "%Y")` and `year(d)`?
4. What does `with_tz()` change?
5. Which function snaps a date to the first day of its month?

Answers

1. `parse_date()`. `parse_datetime()` wants a time as well. `parse_time()` is only a clock time. `ymd_hms()` is lubridate, not readr.

2. `dmy()`. The pieces are day, month, year.
3. `format()` returns a character string. `year()` returns a number. The printed year can look the same.
4. The clock. The instant stays the same, and it is displayed in the other zone. `force_tz()` is the one that keeps the clock and changes the instant.
5. `floor_date(d, unit = "month")`.

---

13. Practice

1. Parse `"2010-02-01"`, `"February 1, 2010"`, `"01-02-2010"`, and `"2010-02-01 08:15:00"`. `"01-02-2010"` is `dmy` or `mdy` depending on the country.
2. Build dates from this tribble and keep the rows before `2010-02-01`.

```
events <- tribble(
  ~year, ~month, ~day,
  2009,     12,   31,
  2010,      1,   15,
  2010,      2,    1,
  2010,      3,   10
)
```

Base R supplies the classes. readr parses on the way in and reports the failures. lubridate is how you work with the dates after that.